

1. Introduction to Data Communication

1.1 Basics of the Communications :

Communication is a process of exchanging information or data between two or more entities through a medium.

In data communication, the data is transmitted in digital form from a sender to a receiver.

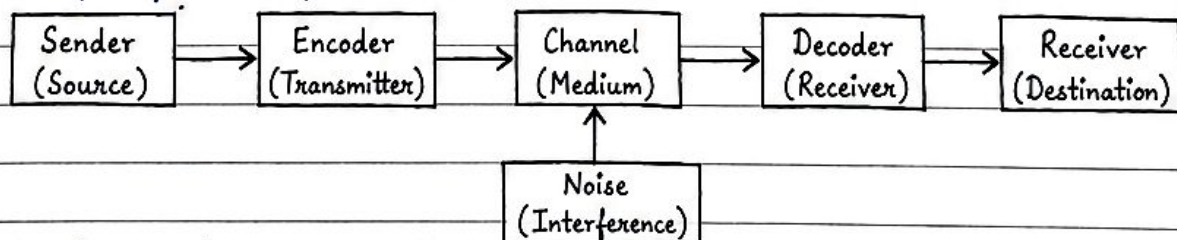
1.1.1 Elements of Communication

A basic communication system consists of the following five essential elements :

- ① Sender / Source - The device that generates the data to be transmitted.
- ② Receiver / Destination - The device that receives the data.
- ③ Message - The information or data to be transmitted.
- ④ Transmission Medium - The physical path through which the data travels.
- ⑤ Protocol - A set of rules that govern the communication process.

1.1.2 Communication Model

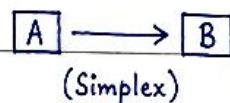
The communication process can be represented with the help of a simple model.



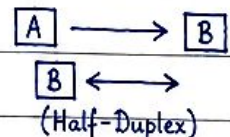
1.1.3 Types of Communication

- ① Simplex Communication - Data flows in only one direction.

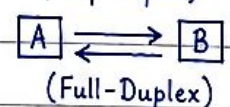
Example: Keyboard to CPU.



- ② Half-Duplex Communication - Data flows in both directions, but not at the same time. Example: Walkie-talkie.



- ③ Full-Duplex Communication - Data flows in both directions simultaneously. Example: Telephone.

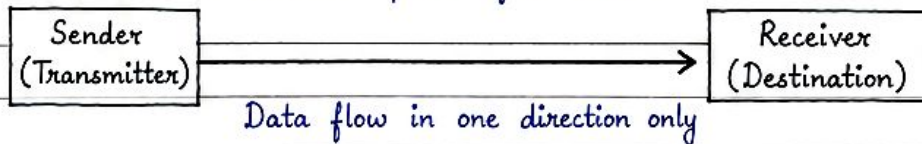


○ Direction of the Data flow

The direction of data flow refers to the way data travels between the sender and receiver in a communication system. It can be in one direction, both directions but not simultaneously, or both directions simultaneously.

① Simplex Transmission :

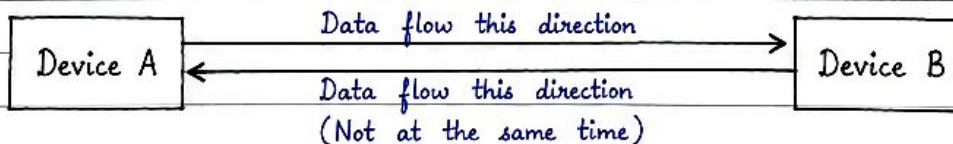
In simplex transmission, data flows in only one direction from sender to receiver. The sender can only transmit and the receiver can only receive. There is no reverse path for data.



Example : Keyboard to CPU, TV Broadcast.

② Half-Duplex Transmission :

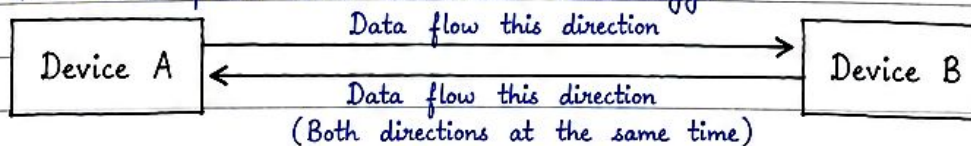
In half-duplex transmission, data can flow in both directions, but not at the same time. When one device is sending, the other can only receive, and vice-versa. It uses the same communication channel.



Example : Walkie-talkie, CB Radio.

③ Full-Duplex Transmission :

In full-duplex transmission, data can flow in both directions simultaneously. Both devices can send and receive data at the same time using separate paths (or same path with advanced technology).



Example : Telephone, Video Call, Internet.

Comparison Table

Type	Direction	Communication	Example
Simplex	One way only	One device sends, other receives	TV Broadcast
Half-Duplex	Both ways, not at same time	One device sends while other receives	Walkie-talkie
Full-Duplex	Both ways, simultaneously	Both devices send and receive together	Telephone, Video Call

○ Network Topologies

The term topology refers to the physical or logical arrangement of nodes and connections in a network.

A network topology defines the way devices are connected and how data flows between them.

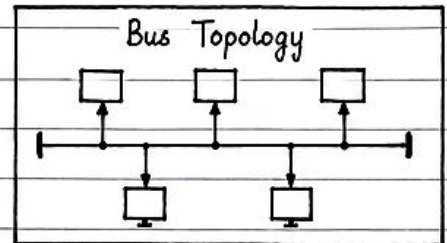
Choosing the right topology is important for performance, reliability, and cost.

① Bus Topology:

In bus topology, all devices are connected to a single central cable called backbone. Data travels in both directions along the bus.

Advantages - Easy to install, requires less cable.

Disadvantages - If backbone fails, entire network fails, difficult to detect faults.

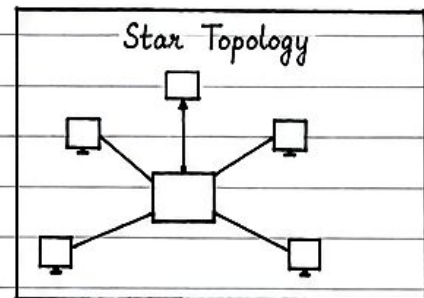


② Star Topology:

In star topology, each device is connected to a central device called hub or switch. All data passes through the central device.

Advantages - Easy to manage, failure of one device does not affect others.

Disadvantages - If central device fails, whole network stops.



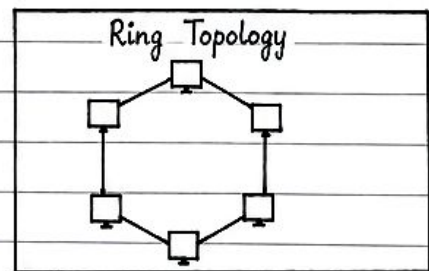
③ Ring Topology:

In ring topology, each device is connected to two other devices, forming a circular path.

Data travels in one direction around the ring.

Advantages - Easy to install and expand, no collisions.

Disadvantages - If one device fails, entire network may be affected.

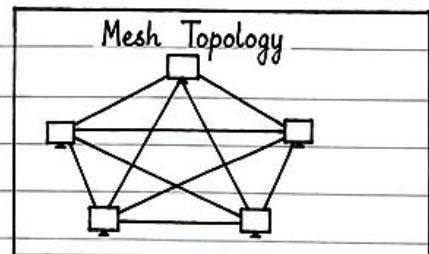


④ Mesh Topology:

In mesh topology, each device is connected to every other device in the network.

Advantages - Highly reliable, data can take multiple paths.

Disadvantages - Expensive, complex cabling installation.

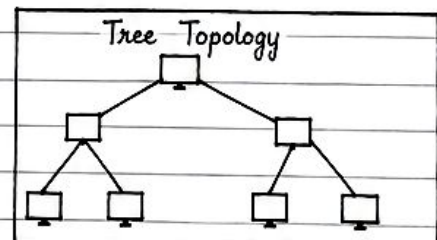


⑤ Tree Topology:

Tree topology is a combination of star and bus topology. It has a central root node and branches to other nodes.

Advantages - Scalable, easy to manage.

Disadvantages - If root node fails, large part of network fails.



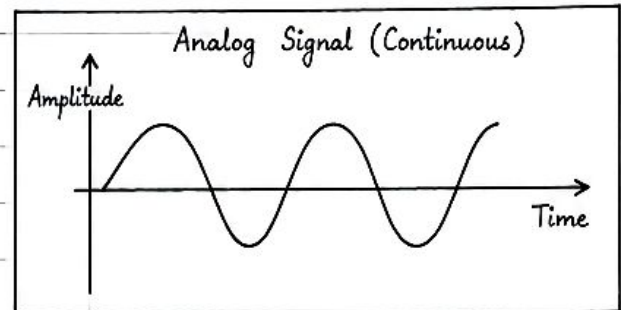
○ Signals and Transmission

In data communication, information is sent from one device to another using signals over a transmission medium.

Signals can be of two types: Analog and Digital.

① Analog Signal :

An analog signal is a continuous electrical waveform that represents data by varying its amplitude, frequency or phase continuously. It can take any value within a range.



Smooth and continuous waveform

Example - Human voice, sound waves, temperature signal.

Advantages :

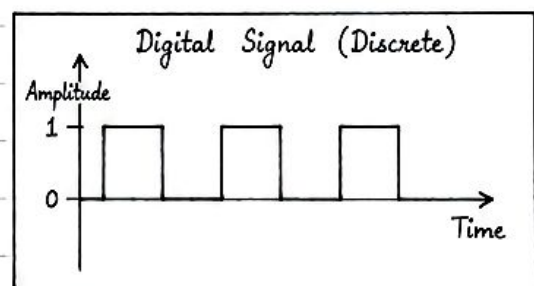
- Can represent real-world data more naturally.
- Suitable for voice, video, and audio signals.

Disadvantages :

- More affected by noise and distortion.
- Regeneration of signal is difficult over long distances.

② Digital Signal :

A digital signal is a discrete signal that represents data in the form of binary digits (0 and 1). It has two distinct levels - HIGH (1) and LOW (0).



Step-like waveform with two levels

Example - Computer data, digital telephone, text, images.

Advantages :

- Less affected by noise.
- Easy to store, process and transmit.
- Signals can be regenerated easily.

Disadvantages :

- Requires more bandwidth.
- Conversion from analog source is required.

★ Comparison between Analog and Digital Signals

Feature	Analog Signal	Digital Signal
Nature	Continuous	Discrete
Signal Level	Infinite levels	Two levels (0 and 1)
Noise Immunity	Low	High
Transmission Quality	Degrades over distance	Remains same, can be regenerated
Bandwidth	Lower	Higher
Examples	Voice, Music, Video	Computer data, Text, Images

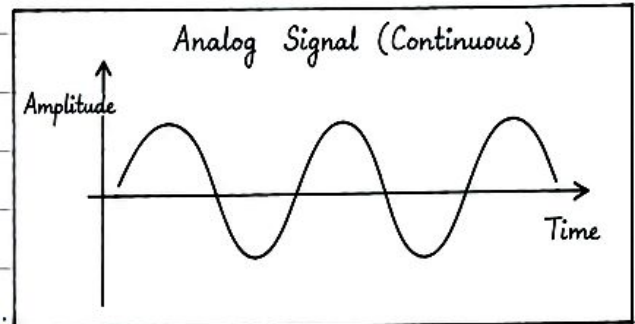
○ Signals and Transmission

In data communication, information is sent from one device to another using signals over a transmission medium.

Signals can be of two types: Analog and Digital.

① Analog Signal

An analog signal is a continuous electrical waveform that varies continuously with time. It can take any value within a range and is used to represent information.



Smooth and continuous waveform

Example - Human voice, sound waves, temperature signal, video signal.

Advantages :

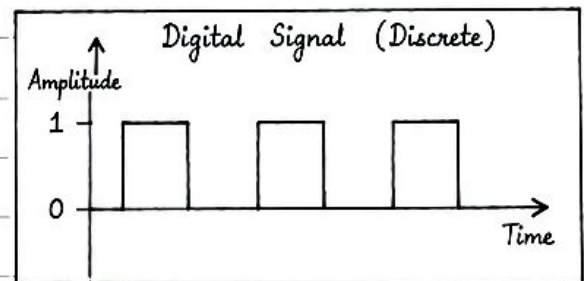
- Can represent real-world data more naturally.
- Suitable for voice, video and audio signals.

Disadvantages :

- More affected by noise and distortion.
- Regeneration of signal is difficult over long distances.

② Digital Signal

A digital signal is a discrete signal that represents data in the form of binary digits (0 and 1). It has two distinct levels - HIGH (1) and LOW (0).



Step-like waveform with two levels

Example - Computer data, digital telephone, text, images.

Advantages :

- Less affected by noise.
- Easy to store, process and transmit.
- Signals can be regenerated easily.

Disadvantages :

- Requires more bandwidth.
- Conversion from analog source is required.

★ Comparison between Analog and Digital Signals

Feature	Analog Signal	Digital Signal
Nature	Continuous	Discrete
Signal Level	Infinite levels	Two levels (0 and 1)
Noise Immunity	Low	High
Transmission Quality	Degrades over distance	Remains same, can be regenerated
Bandwidth	Lower	Higher
Examples	Voice, Music, Video	Computer data, Text, Images

○ Transmission Media

Transmission media is the physical path through which data travels from the sender to the receiver. It acts as a channel to carry signals between devices.

Transmission media can be classified into two types :

Guided (Wired) and Unguided (Wireless).

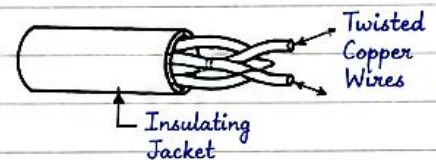
1. Guided Transmission Media (Wired)

Guided media use physical cables or wires to transmit data. Signals travel in a specific path and are not affected by external atmospheric conditions.

Types :

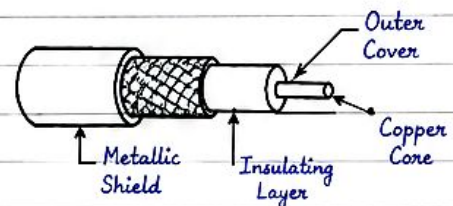
① Twisted Pair Cable

- Consists of two insulated copper wires twisted together.
- Used in telephone lines and LANs.
- Example: Ethernet cable.



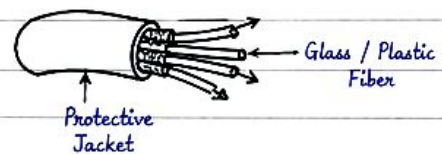
② Coaxial Cable

- Has a central copper conductor surrounded by an insulating layer, metallic shield, and outer protective cover.
- Used in cable TV, broadband internet.



③ Optical Fiber Cable

- Made of thin strands of glass or plastic that transmit data as light pulses.
- High speed, high bandwidth and low signal loss.
- Used in internet backbone, telecommunication.



2. Unguided Transmission Media (Wireless)

Unguided media transmit signals through air or space without using any physical cable. Signals are broadcast in all directions and can be received by many devices.

Types :

① Radio Waves

- Used in radio broadcasting, TV, mobile communication.
- Can penetrate walls and buildings.



Radio Tower

② Microwave

- High frequency radio waves used for communication.
- Used in satellite links, Wi-Fi, GPS.



Microwave Dish

③ Infrared Waves

- Short range communication using infrared light.
- Used in remote controls, IR data transfer.



IR Remote

★ Comparison between Guided and Unguided Media

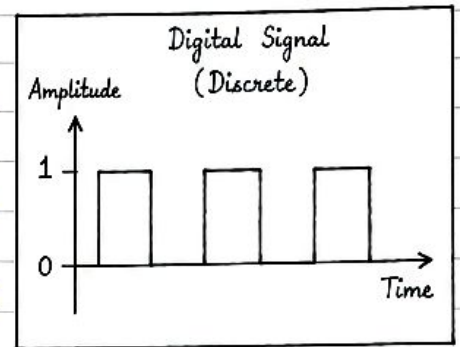
Feature	Guided (Wired)	Unguided (Wireless)
Medium	Physical cables or wires	Air or space
Signal Direction	One specific path	All directions
Interference	Less affected	More affected by environment
Security	More secure	Less secure
Installation Cost	Higher	Lower
Examples	Twisted Pair, Coaxial, Fiber Optic	Radio, Microwave, Infrared

○ Concept of Digital Signals, Bit rate, Bit length

1. Concept of Digital Signals:

A digital signal is a discrete signal that can take only a finite number of distinct values. In computer networks, digital signals are usually represented in binary form, i.e., 0 and 1. These signals are used to transmit data over communication media.

They are less affected by noise and can be regenerated easily, making them more reliable than analog signals.



Digital signal waveform

2. Bit Rate:

Bit rate refers to the number of bits transmitted per second from sender to receiver. It is measured in bits per second (bps) or its multiples like Kbps (10^3 bps), Mbps (10^6 bps), Gbps (10^9 bps). Higher bit rate means more data can be transmitted in a given time.

Formula:

$$\text{Bit Rate} = \frac{\text{Total number of bits}}{\text{Total time (in seconds)}}$$

Unit: bits per second (bps)

Example: If 2,000,000 bits are transmitted in 2 seconds,

$$\text{then bit rate} = \frac{2,000,000}{2} = 1,000,000 \text{ bps} = 1 \text{ Mbps.}$$

3. Bit Length:

Bit length is the duration of a single bit. It is the time required to transmit one bit from sender to receiver. It is measured in seconds (s) or microseconds (μs).

Formula:

$$\text{Bit Length} = \frac{\text{Total time}}{\text{Total number of bits}}$$

Relationship between Bit Rate and Bit Length

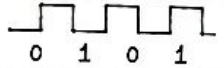
$$\text{Bit Length} = \frac{1}{\text{Bit Rate}}$$

This means, if bit rate increases, bit length decreases and vice versa.

Example: If bit rate = 1 Mbps (1,000,000 bps),

$$\text{then bit length} = \frac{1}{1,000,000} = 1 \text{ microsecond } (\mu\text{s})$$

Summary Table

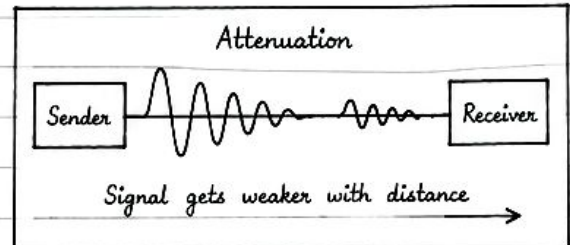
Term	Meaning	Formula	Unit	Example
Digital Signal	Signal that can take only discrete values (0 and 1)	—	—	
Bit Rate	Number of bits transmitted per second	$\frac{\text{Total bits}}{\text{Total time}}$	bps, Kbps, Mbps, Gbps	2,000,000 bits in 2 sec = 1 Mbps
Bit Length	Time taken to transmit one bit	$\frac{\text{Total time}}{\text{Total bits}} = \frac{1}{\text{Bit Rate}}$	seconds (s), microseconds (μs)	At 1 Mbps, bit length = 1 μs

○ Transmission Impairment

When a signal travels from sender to receiver through a transmission medium, it gets affected by various unwanted effects called transmission impairments. These impairments degrade the signal and may cause distortion, loss of data or errors. The main impairments are Attenuation, Distortion, Noise and Delay.

① Attenuation :

Attenuation is the loss of signal strength as it travels through the transmission medium. The signal becomes weaker with distance. If attenuation is high, the signal may become too weak to be detected at the receiver.

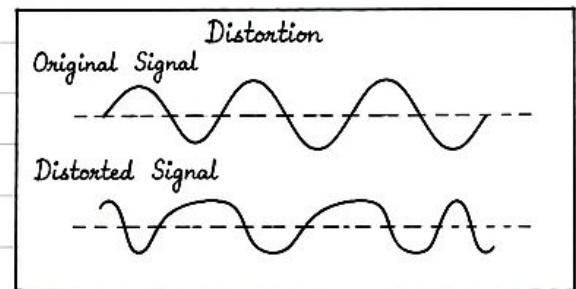


Example : Signal loss in long distance communication.

Effect : Reduced signal strength, may cause data loss.

② Distortion (Signal Distortion) :

Distortion occurs when different components of a signal travel at different speeds or get delayed by the medium. As a result, the shape of the signal changes and the received signal becomes different from the original signal.

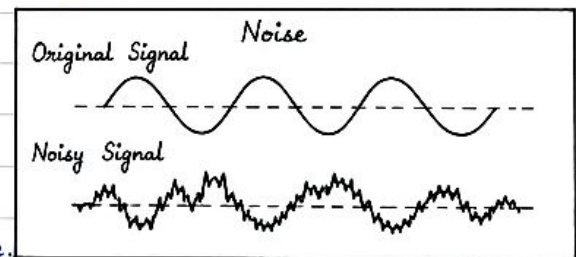


Example : Distortion in audio signal over long distance.

Effect : Signal becomes unclear, affecting data accuracy.

③ Noise :

Noise is any unwanted signal that mixes with the original signal and interferes with its transmission. Noise is generated by external sources or by the electronic components of the system.

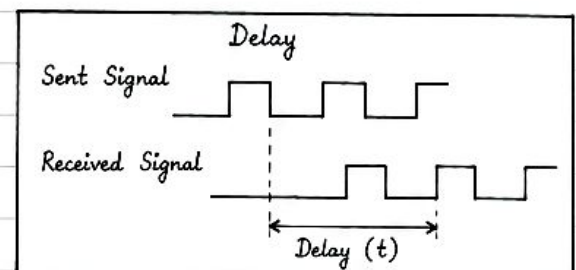


Example : Electrical noise, cross-talk, thermal noise.

Effect : May corrupt the signal and cause errors.

④ Delay (Time Delay) :

Delay is the time taken by the signal to travel from sender to receiver. Excessive delay can cause synchronization problems, especially in real-time communication.



Example : Delay in satellite communication.

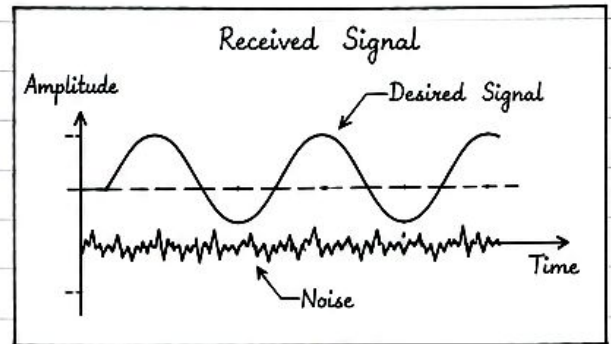
Effect : May cause late delivery, affecting real-time communication.

Comparison of Transmission Impairments

Impairment	Cause	Effect	Example
Attenuation	Loss of signal strength in medium	Weak signal, may cause data loss	Signal loss in long cable
Distortion	Different components travel at different speeds	Change in signal shape, less accurate	Audio distortion
Noise	Unwanted signals mix with original signal	Signal corruption, errors occur	Electrical noise, cross-talk
Delay	Time taken by signal to reach receiver	Late delivery, sync problems	Satellite communication delay

○ Signal to Noise Ratio (SNR)

Signal to Noise Ratio (SNR) is a measure used to compare the level of a desired signal to the level of background noise. It indicates how strong the useful signal is in comparison to the unwanted noise. A higher SNR means better quality of communication, while a lower SNR means more distortion and errors in the received signal.



Received signal with noise

1. What is Signal to Noise Ratio ?

SNR is the ratio of signal power to noise power. It is usually expressed in decibels (dB). It helps in determining the clarity and reliability of a communication system.

2. Formula :

$$\text{SNR} = \frac{\text{Signal Power } (P_s)}{\text{Noise Power } (P_n)}$$

or

$$\text{SNR (dB)} = 10 \log_{10} \left(\frac{P_s}{P_n} \right)$$

Where,

P_s = Power of desired signal

P_n = Power of noise

SNR(dB) = Signal to Noise Ratio in decibels

Decibel (dB) :

Decibel is a logarithmic unit used to express the ratio of two power values.

$$\text{dB} = 10 \log_{10} (\text{Power ratio})$$

3. Interpretation of SNR :

- High SNR (e.g., > 30 dB) → Strong signal, low noise, high quality.
- Moderate SNR (e.g., 10 - 30 dB) → Acceptable quality.
- Low SNR (e.g., < 10 dB) → Weak signal, high noise, poor quality.

4. Importance of SNR :

- Determines the quality and reliability of data transmission.
- Helps in reducing errors in digital communication.
- Used in designing communication systems, receivers and amplifiers.
- Higher SNR allows higher data rates and better performance.

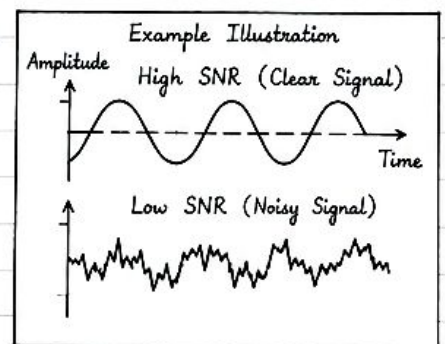
5. Example :

If the signal power = 100 mW and noise power = 1 mW,

$$\text{then SNR} = \frac{100}{1} = 100$$

$$\text{SNR (dB)} = 10 \log_{10} (100) = 20 \text{ dB}$$

This means the signal is 20 dB stronger than the noise.



★ Factors affecting SNR :

Factor	Effect on SNR
Increase in signal power	Increases SNR (better quality)
Increase in noise power	Decreases SNR (poor quality)
Bandwidth	Wider bandwidth may increase noise, reducing SNR
Transmission medium	Better medium gives higher SNR
Interference	More interference lowers SNR

Summary

SNR is a key parameter in communication systems. It tells us how much the useful signal stands out from the background noise. For good communication, we should aim for higher SNR.